Byeonghun Lee

Ph.D. Student, School of Electrical Engineering, Korea University
145 Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea
byeonghun_lee@korea.ac.kr | +82) 10-4608-7133

EDUCATION

Mar. 2022 – Present	**Korea University** School of Electrical Engineering Ph.D. Student (Advisor: Kyong Hwan Jin)	Seoul, Korea
Mar. 2017 – Feb. 2022	**Daegu Gyeongbuk Institute of Science and Technology (DGIST)** School of Undergraduate Studies B.S. in Convergence	Daegu, Korea

RESEARCH INTERESTS

- Image Processing
- Inverse Problems
- Retrieval-Augmented Generation in Vision
- Image Super-Resolution
- 3D Rendering, Depth Estimation

PUBLICATIONS

1. **Byeonghun Lee**, Jaedeok Kim, Sunghoon Im and Kyong Hwan Jin, "Target-Oriented Neural MPI Stitching for Novel View Synthesis," *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027*.
2. **Byeonghun Lee**, Hyunmin Cho, Sunghoon Im†, and K. H. Jin†, "ReAL: Reference-to-Image (R2I) Aware Latent Diffusion for Image Super-Resolution," *European Conference on Computer Vision (ECCV), 2026*.
3. **Byeonghun Lee***, Hyunmin Cho*, Honggyu Choi, Soo Min Kang, Iljun Ahn and K. H. Jin*, "Reference-based Super-Resolution via Image-based Retrieval-Augmented Generation Diffusion," *IEEE/CVF International Conference on Computer Vision (ICCV), 2025*.
4. Woo Kyoung Han*, Yongjun Lee*, **Byeonghun Lee**, Sang Hyun Park, Sunghoon Im*, and K. H. Jin*, "JPEG Processing Neural Operator for Backward-Compatible Coding," *IEEE/CVF International Conference on Computer Vision (ICCV), 2025*.
5. Woo Kyoung Han, **Byeonghun Lee**, Hyunmin Cho, Sunghoon Im*, K. H. Jin*, "Towards Lossless Implicit Neural Representation via Bit Plane Decomposition," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025*.
6. EungGu Kang, **Byeonghun Lee**, Sunghoon Im*, and K. H. Jin*, "BurstM: Deep Burst Multi-scale SR using Fourier Space with Optical Flow," *European Conference on Computer Vision (ECCV), 2024*.
7. Minsu Kim, Jaewon Lee, **Byeonghun Lee**, Sunghoon Im and K. H. Jin, "Implicit Neural Image Stitching With Enhanced and Blended Feature Reconstruction," *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024*.
8. Woo Kyoung Han, **Byeonghun Lee**, Sang Hyun Park, and K. H. Jin, "ABCD: Arbitrary Bitwise Coefficient for De-quantization," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023*.

PATENTS

1. Apparatus and Method of Recovering Image Using Arbitrary Bitwise Coefficient Estimation for Dequantization, KR 10-2023-0111027 (2023-08-24)
2. Method and Apparatus for Lossless Implicit Neural Representation using Bit-plane Decomposition, KR 10-2025-0151831 (2025-10-20)
3. Polarization-related patents (Samsung Electronics collaboration) — 1 international patent application, pending

AWARDS AND HONORS

Bronze Medal, Paper Award — IPIU, Korea Jun. 2026

RESEARCH PROJECTS

Synthetic Polarization Data Generation from RGB Images & Polarization-Based Precise Depth Estimation — Samsung Electronics, GTR Mar. 2026 – Present

Hallucination Mitigation in Video Understanding Models — Samsung Research Dec. 2025 – Present

User-Preference-Based Image Quality Enhancement — Samsung Research Jun. 2024 – May 2025

Cognitive Reverse-Haptics Reproduction System Lab — Basic Research Laboratory (BRL), National Research Foundation of Korea Feb. 2023 – Feb. 2024

Developed signal-processing algorithms to convert real-world tactile data into digital signals, and designed a Verilog-based hardware demonstration.

Food Classification Algorithm for Caloric Intake Analysis — Pinset Healthcare Aug. 2022 – Jul. 2023

Developed food classification and calorie estimation algorithms from user-captured food images.

Deep Learning Algorithm for Bacteria Detection — THE WAVE TALK Mar. 2022 – Aug. 2022

Developed an image-based classification algorithm to identify bacteria species and concentration levels.

SKILLS

- Background in 3D rendering and image warping
- Expertise in image super-resolution techniques
- Research and development experience with diffusion models
- Expertise in implicit neural representations (INR) for continuous and lossless image representation
- Experience in reference-based super-resolution and retrieval-augmented generation for vision
- Knowledge of image compression and de-quantization
- Experience in polarization imaging and depth estimation